4 Energy transport by a wave

4.1 The energy in a wave

Having established that nothing material is actually transported from one point to another by a wave, it seems strange to think that energy is transported by a wave. But consider again the stadium wave, for example: At a certain time, spectator A goes from sitting still to standing up, so that A's kinetic as well as potential energy are increasing, while spectator B down the row is sitting still. A little bit later, A has gone back to sitting still, and B's energy is changing as he stands up. So we can legitimately say that energy has been carried from A's location to B's.

Similarly, a transverse wave propagating along a string, or a pressure wave in a gas, carry energy from one location to another. In the next section, we explore the transport of energy by a wave quantitatively.

4.2 The energy transported by a transverse wave in a string

Consider for example a very small segment of a vibrating string of length δx at a position x along the string, as illustrated in Figure 4.1. As the string is vibrating, the segment is moving with vertical velocity $\partial y/\partial t$, and therefore has kinetic energy

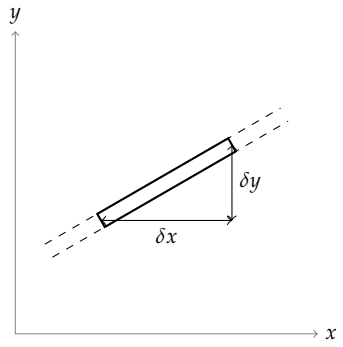


Figure 4.1: A small segment of a vibrating string.

$$\delta K = \frac{1}{2}\mu\delta x\left(\frac{\partial y}{\partial t}\right)^2 \tag{4.1}$$

where μ is the mass per unit length of the string, so the mass of the segment is $\mu\delta x$.

As for the potential energy, we should note that the equilibrium length of the segment when the string is not vibrating is δx. But as the string vibrates the segment is stretched to a length

$$\delta s = \sqrt{\delta x^2 + \delta y^2} \tag{4.2}$$

We can assume that $\delta y \ll \delta x$, so

$$\delta s \approx \delta x\sqrt{1+\left(\frac{\partial y}{\partial x}\right)^2} \approx \delta x\left[1+\frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2\right] \tag{4.3}$$

Thus the string stretches by an amount $\delta s - \delta x$, and the potential energy changes by an amount equal to the work done by the tension T, which we can assume to be constant. Setting $U = 0$ when the elongation of the segment is zero, we can write

$$\delta U = T(\delta s - \delta x) = \frac{1}{2} T \delta x \left(\frac{\partial y}{\partial x}\right)^2 \tag{4.4}$$

The total mechanical energy of the segment is therefore

$$\delta E = \delta K + \delta U = \frac{1}{2}\mu \delta x \left(\frac{\partial y}{\partial t}\right)^2 + \frac{1}{2} T \delta x \left(\frac{\partial y}{\partial x}\right)^2 \tag{4.5}$$

Remembering that the propagation speed of the wave is $v = \sqrt{T/\mu}$, we can write the energy as

$$\delta E = \frac{1}{2} T \delta x \left[\frac{1}{v^2}\left(\frac{\partial y}{\partial t}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2\right] \tag{4.6}$$

4.3 *Power transported by a transverse wave in a string*

As equation (4.6) shows, the energy of the segment is proportional to its length δx, so the energy *per unit length* of the segment is

$$\frac{\delta E}{\delta x} = \frac{1}{2} T \left[\frac{1}{v^2}\left(\frac{\partial y}{\partial t}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2\right] \tag{4.7}$$

Now let's consider how much energy is flowing past a point in the string during a time dt. Since the wave is passing the point with speed v, during that time a length $v\,dt$ passes by the point, and it carries an amount of energy equal to

$$dE = \frac{\delta E}{\delta x} \times v\,dt \tag{4.8}$$

So the energy flowing past the point per second, i.e. the *power* transported by the wave at that point, is

$$P = \frac{dE}{dt} = \frac{\delta E}{\delta x} v = \frac{1}{2} v T \left[\frac{1}{v^2}\left(\frac{\partial y}{\partial t}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2\right] \tag{4.9}$$

We can simplify this even further by taking into account the fact that for a wave, the derivatives $\partial y/\partial t$ and $\partial y/\partial x$ are related: Since $y = f(x - vt)$,

$$\frac{\partial y}{\partial t} = -v f'(x - vt) \tag{4.10a}$$

$$\frac{\partial y}{\partial x} = v f'(x - vt) \tag{4.10b}$$

so

$$P = vT[f'(x - vt)]^2 \tag{4.11}$$

4.4 A sinusoidal transverse wave in a string

As an example, let's look at the particular case of a sinusoidal wave. As we have seen, we can write the vertical displacement of the string y at a point x as

$$y = A\cos[k(x - vt)] \tag{4.12}$$

In this case the expression for the power transmitted by the wave is

$$P = k^2 vTA^2 \sin^2(kx - \omega t) \tag{4.13}$$

or, since $\omega = kv$,

$$P = \omega^2 \frac{T}{v} A^2 \sin^2(kx - \omega t) \tag{4.14}$$

Now at any given position x this is a function of time, so we must call this the **instantaneous power**. The average of the $\sin^2$ function over time is actually exactly equal to $1/2$, so the **average power** transmitted by this wave at any point is

$$P_{av} = \frac{1}{2}\omega^2 \frac{T}{v} A^2 \tag{4.15}$$

We note that the average power is proportional to the frequency squared, and to the amplitude squared. This is a significant fact, that we shall make use of later.

4.5 The mechanical impedance and the power transported by a generic wave

Definition of mechanical impedance

Previously, when we were deriving the expression for the speed of a transverse wave in an elastic string, we found that the transverse force was related to the slope of the string by the expression of equation (3.2):

$$F_y = -T\frac{\partial y}{\partial x} \tag{4.16}$$

which for a wave $y = f(x - vt)$ is equivalent to

$$F_y = -Tf'(x - vt) \tag{4.17}$$

At the same time,

$$v_y = \frac{\partial y}{\partial t} = -vf'(x - vt) \tag{4.18}$$

so that if we construct the ratio $Z = F_y/v_y$, we get a parameter that is independent of the type of wave, and only depends on properties of the medium itself (in this case, the tension in an elastic string):

$$Z = \frac{F_y}{v_y} = \frac{T}{v} \tag{4.19}$$

This is called the **mechanical impedance** of the medium. Note, by the way, that because $v = \sqrt{T/\mu}$, an alternative expression for the impedance of an elastic string is

$$Z = \sqrt{\mu T} \tag{4.20}$$

What does it mean to have a large or a small value of the impedance? Roughly speaking, F_y is the force that causes the medium to vibrate with velocity v_y. If the ratio is large, we can say that a large force produces only a small perturbation. The medium is not wave-friendly! Whereas if the ratio is small, a small force produces a large perturbation, and the medium offers easy passage to a wave.

Why bother to define Z?

Here's why this is such a useful quantity: If we substitute Z in the expressions for the instantaneous and average power of a wave in a string, eqs. (4.14) and (4.15), they become

$$P = \omega^2 Z A^2 \sin^2(kx - \omega t) \tag{4.21}$$

and

$$P_{av} = \frac{1}{2}\omega^2 Z A^2 \tag{4.22}$$

Z contains all the information about the particular medium that carries the wave. The expressions for the power are quite general, and could be applied to a sinusoidal wave in any medium, simply by substituting the appropriate expression for Z in that medium.

Example 4.1: *A wave propagates along a string under tension T, with frequency ω, wavelength λ, and amplitude A. If the tension is doubled, while the wavelength of the wave remains the same, how does the speed v change? How does the frequency of the wave change? How does the mechanical impedance of the string change? If the average power transmitted by the wave remains the same, how does the amplitude change?*

Since $v = \sqrt{T/\mu}$, doubling the tension implies that the speed increases by a factor $\sqrt{2}$.

The frequency $\omega = kv$ and since the wavelength remains the same, ω increases by the same factor as the speed.

The mechanical impedance is $Z = \sqrt{\mu T}$, so doubling the tension implies that Z also increases by a factor $\sqrt{2}$.

Then from equation (4.22) we see that the amplitude is reduced by a factor

$$\sqrt{\frac{1}{2\sqrt{2}}} = 0.60$$

Transport of energy by longitudinal waves in a fluid – Acoustic impedance

Consider now a sound wave in a fluid column. In the earlier section "The speed of sound in a fluid" we saw that the net pressure on a thin layer of fluid of cross-sectional area S[1] could be put in terms of the width of the layer δx, and the longitudinal displacement of the layer δy, with the help of the definition of bulk modulus B[2]

[1] We won't call the area A this time, to avoid confusion with the amplitude of a wave a little later.
[2] See equation (3.10).

$$\delta p = -B\frac{\delta y}{\delta x} \tag{4.23}$$

and consequently the longitudinal force on the layer is

$$F_y = \delta p A = -BS\frac{\delta y}{\delta x} \tag{4.24}$$

If we let the layer be infinitesimally thin,

$$F_y = -BS\frac{\partial y}{\partial x} \tag{4.25}$$

Again, the velocity of the layer as it moves under the action of the wave is $v_y = \partial y/\partial t$, and because the displacement y is a function $f(x - vt)$, it still holds true that

$$\frac{\partial y}{\partial x} = f'(x - vt) \tag{4.26a}$$

$$\frac{\partial y}{\partial t} = vf'(x - vt) \tag{4.26b}$$

$$\tag{4.26c}$$

Now we can construct the mechanical impedance for a fluid column as

$$Z = \left|\frac{F_y}{v_y}\right| = \frac{BS}{v} \tag{4.27}$$

In this expression B and v are properties of the medium, but the cross-sectional area S is still a property of a particular column, so it's best to work with the mechanical impedance per unit area, which we will call the **acoustic impedance** of the medium:

$$Z_a = \frac{B}{v} \tag{4.28}$$

Another useful expression for the acoustic impedance, taking into consideration that $v = \sqrt{B/\rho}$, is

$$Z_a = \sqrt{B\rho} \tag{4.29}$$

Now the power transported by the wave is again given by equations (4.21) and (4.22), but because the *mechanical impedance* Z still depends on S, it is best to work with the power per unit cross-sectional area, which is called the **intensity** of the sound wave I. The instantaneous intensity thus turns out to be

$$I = \omega^2 \frac{B}{v} A^2 \sin^2(kx - \omega t) \tag{4.30}$$

and the average intensity

$$I = \frac{1}{2}\omega^2 \frac{B}{v} A^2 \tag{4.31}$$

Example 4.2: *Calculate the acoustic impedance of air at 1 atm of pressure, with density* $\rho = 1.3\,\mathrm{kg\,m^{-3}}$.

The bulk modulus for air is γp, where $\gamma = 7/5$ for diatomic molecules. With these values,

$$Z_{air} = \sqrt{B\rho} = 429\,\mathrm{kg\,m^{-2}\,s^{-1}}$$

4.6 *Energy transported by waves propagating in 2 and 3 dimensions*

Why do sounds become fainter with increasing distance to the source?

First, we would have to investigate what "fainter" (or "louder") means. The key component in the ear for sensing sound is the **eardrum**, a thin membrane capable of oscillating. When a sound wave reaches the eardrum, it vibrates with an amplitude that depends on the amount of energy it absorbs from the wave. The larger the amplitude with which the eardrum vibrates, the "louder" the sound, according to the brain.

The amount of energy reaching the eardrum each second is equal to the intensity of the wave, which is the power transported by the wave per unit area, multiplied by the area of the eardrum. So the loudness of the sound is a measure of the intensity of the wave. We will quantify the perceived loudness of a sound very shortly, but for now let's consider the intensity of the wave that reaches an observer at a certain distance from the source.

Assume to begin with a point source of sound that emits uniformly in all directions with power P. Also, ignore for the moment the possible

attenuation of the sound as it propagates through the air. (We will deal with that a little later, too). Each spherical wavefront that spreads out from the source carries this power. But the surface area of a spherical wavefront of radius r is $4\pi r^2$, so the intensity, or power per unit area, at a point a distance r from the source is $P/(4\pi r^2)$. This is the intensity that reaches the eardrum of an observer at a distance r. Since the intensity is inversely proportional to the distance squared – a result known as the **inverse square law** – it follows that the loudness of the sound decreases with distance from the source.

If the source is a line (e.g. a vibrating wire) emitting uniformly in all directions, the wavefronts are cylindrical, as we saw in Figure 2.6. The surface area of a cylinder of length L and radius r is $2\pi rL$, so the intensity at a distance r would be $P/(2\pi rL)$, which is inversely proportional to r. So a line source does not follow an inverse square law. The intensity still drops off with distance, but a slower rate $(1/r)$ than for a point source $(1/r^2)$.

In both cases, the intensity drops because the surface area of a wavefront increases with distance from the source. We call this phenomenon **geometric spreading**.

As for a plane source of sound, if we look again at Figure 2.6, we see that the surface area of the wavefronts is constant. So in this case, the intensity would be the same at any distance! Of course, strictly this would be true only for an infinite plane source. If the source is finite in size, at very large distances it looks more like a point, and the inverse square law applies again. But at small distances compared with the size of the source, it is true that the intensity is very close to being independent of distance.

As we mentioned above, another reason why the intensity could drop off with distance is **attenuation**. The medium can absorb some energy from the wave as it passes. For a plane wave, with no geometrical spreading, the intensity nevertheless decreases exponentially as

$$I = I_u e^{-\alpha x} \tag{4.32}$$

where I_u is the unattenuated intensity, and α is the **attenuation coefficient** of the medium, with dimensions of inverse length. We assume that α itself is constant, which is a good approximation for our purposes.

Similarly, for a spherical or a cylindrical wave,

$$I_{sph} = \frac{P}{4\pi r^2} e^{-\alpha r} \tag{4.33a}$$

$$I_{cyl} = \frac{P}{2\pi rL} e^{-\alpha r} \tag{4.33b}$$

Example 4.3: *A point source of sound emits* 20 W. *What is the unattenu-*

ated intensity at 100 m*? If the attenuation coefficient is* $\alpha = 0.01\,\mathrm{m}^{-1}$*, what is the intensity at* 100 m*?*

The unattenuated intensity is

$$I_u = \frac{P}{4\pi r^2} = 1.59 \times 10^{-4}\,\mathrm{W\,m}^{-2}$$

The attenuated intensity is

$$I = I_u e^{-\alpha r} = 0.3679 I_u = 5.85 \times 10^{-5}\,\mathrm{W\,m}^{-2}$$

4.7 *The perception of sound: Acoustic intensity level and the decibel scale*

Weber's law

After a physical discussion of the transport of energy by waves, and in particular sound waves, it is natural to investigate whether there is any relation between our perception of loudness, and the energy content of a sound wave. It turns out there is, but as one might expect given the complexity of our sensory processes, the relation is not simple.

Experimental psychologists have been studying the relation between the magnitude of a stimulus to the senses, and the magnitude of the our response to the stimulus for 200 years. In the early 1800's, Ernst Weber conducted experiments in what came to be known as the science of **psychophysics**. In one experiment, he put a weight S on a blindfolded subject's hand[3], and added small weights to it, to determine the smallest weight difference dS that the subject could perceive. He found that the smallest *perceived* weight difference dp was in fact proportional to the *ratio* of the actual weight difference dS to the original weight S:

[3] S stands for "stimulus" here. The results of these experiments can be extended to other types of stimuli, including sound.

$$dp = k\frac{dS}{S} \tag{4.34}$$

Integrating,

$$p = k \ln S + C \tag{4.35}$$

where C is an integration constant.

There is generally a threshold value S_0 below which there is no perception. Setting $p = 0$ in equation (4.35) and solving for C, the relation becomes

$$p = k \ln\left(\frac{S}{S_0}\right) \tag{4.36}$$

This result, namely that the perceived measure of a stimulus is proportional to the logarithm of the magnitude of the stimulus, is known as **Weber's law**, and it applies not only to weights, but also to sound and light, within a wide range of magnitudes.

The decibel scale

When we perceive sound, if the stimulus is the *intensity* of the sound, we call the corresponding perception the **sound level** β, which, following Weber's law, is measured on a logarithmic scale:

$$\beta = 10 \log\left(\frac{I}{I_0}\right) \tag{4.37}$$

where I_0 is the **threshold of hearing**, defined[4] to be $I_0 = 10^{-12}\,\mathrm{W\,m^{-2}}$. The use of logarithms to base 10 is chosen with our decimal numbering system in mind, and of course log is proportional to ln. The factor 10 in front is chosen to give typical values between 0 and approximately 100. Notice that a sound level of zero corresponds to the threshold of hearing.

[4] In fact, the actual threshold of hearing depends on frequency, and it also varies with factors like age or gender. So the value of $10^{-12}\,\mathrm{W\,m^{-2}}$ is chosen arbitrarily, although it is close to typical values of the real threshold of hearing.

Although sound level is dimensionless, when it is measured according to equation (4.37), it is said to be measured in **decibels**, abbreviated dB. Table 4.1 shows the decibel values corresponding to various levels of sound.

dB	Sound
0	Threshold of hearing
20	Rustling leaves
30	Quiet whisper at 1 meter
40	Quiet home
50	Quiet street
60	Normal conversation
70	Inside a car
75	Loud singing at 1 meter
80	Car at 8 meters
88	Motorcycle at 10 meters
90	Blender at 1 meter
100	Diesel truck at 10 meters
107	Power mower at 1 meter
115	Pneumatic riveter at 1 meter
117	Chainsaw at 1 meter
120	Amplified rock and roll music at 2 meters
130	Jet plane at 30 meters

Table 4.1: Typical decibel values for different levels of sound

Example 4.4: *According to table 4.1, a Diesel truck at 10 meters produces 100 dB of sound. What is the acoustic power of the engine, assuming it can be treated as a point source?*

$\beta = 100$ dB corresponds to an intensity $I = 1.00 \times 10^{-2}\,\mathrm{W\,m^{-2}}$. Then

$$P = 4\pi r^2 I = 12.6\,\mathrm{W}$$

Some consequences of Weber's law

The fact that our perception of sound is logarithmic, rather than linear, leads to some interesting effects:

- Doubling the intensity is not equivalent to doubling the intensity level! In fact, when the intensity of a sound is doubled, its intensity

level rises by a fixed amount. Let's say sound #1 has an intensity I_1, and consequently an intensity level

$$\beta_1 = 10 \log\left(\frac{I_1}{I_0}\right) \tag{4.38}$$

Sound #2 has an intensity $I_2 = 2I_1$, so its intensity level is

$$\begin{aligned} \beta_2 &= 10 \log\left(\frac{I_2}{I_0}\right) \\ &= 10 \log\left(\frac{2I_1}{I_0}\right) \\ &= 10 \log(2) + 10 \log\left(\frac{I_1}{I_0}\right) \\ &= 3.01 + \beta_1 \end{aligned} \tag{4.39}$$

So a doubling of the intensity results in an additional 3.01 dB in the intensity level.

- As we have seen, the intensity of a sound produced by a distant source, varies with the distance to the source. As a consequence, the intensity level also varies with distance. Let's compare the intensity level β_1 at a distance r from a source of power P, and the intensity level β_2 at twice the distace from the source, $2r$. Ignoring attenuation, we have

$$\beta_1 = 10 \log\left(\frac{P}{4\pi r^2 I_0}\right) \tag{4.40a}$$

$$\beta_2 = 10 \log\left(\frac{P}{4\pi (2r)^2 I_0}\right) \tag{4.40b}$$

Therefore

$$\beta_1 - \beta_2 = 10 \log(4) = 6.02 \tag{4.41}$$

Example 4.5: *An observer is standing exactly half way between two identical point sources of sound, and hears a sound level β_1. She then moves to a point twice as close to one of the sources as to the other, and hears a sound level β_2. What is the difference $\beta_2 - \beta_1$?*

This example is remarkable because of the scarcity of numerical data. Let's call the power of each source P, and the separation between the sources d. In the first case, she hears an intensity

$$I_1 = \frac{2P}{4\pi (d/2)^2} = \frac{2P}{d^2}$$

In the second case, she hears an intensity

$$I_2 = \frac{P}{4\pi(d/3)^2} + \frac{P}{4\pi(2d/3)^2} = \frac{45P}{16\pi d^2}$$

Then

$$\beta_2 - \beta_2 = 10\log\left(\frac{I_2}{I_1}\right) = 10\log\left(\frac{45}{32}\right) = 1.48 \text{ dB}$$